

Course Objectives:

The main learning objective of this course is to prepare the students for:

- ☑ To learn different cipher techniques
- ☑ To implement the algorithms DES, RSA
- ☑ To implement the algorithms MD5, SHA-1
- ☑ To use network security tools
- ☑ To use network vulnerability assessment tools

LIST OF EXPERIMENTS

1. Perform encryption, decryption using the following substitution techniques
(i) Ceaser cipher, (ii) playfair cipher iii) Hill Cipher iv) Vigenere cipher
2. Perform encryption and decryption using following transposition techniques
i) Rail fence ii) row & Column Transformation
3. Apply DES algorithm for practical applications.
4. Apply AES algorithm for practical applications.
5. Implement RSA Algorithm using HTML and JavaScript
6. Implement the Diffie-Hellman Key Exchange algorithm for a given problem.
7. Calculate the message digest of a text using the SHA-1 algorithm.
8. Implement the SIGNATURE SCHEME - Digital Signature Standard.
9. Demonstrate intrusion detection system (ids) using any tool eg. Snort or any other s/w.
10. Automated Attack and Penetration Tools
Exploring N-Stalker, a Vulnerability Assessment Tool
11. Defeating Malware
i) Building Trojans ii) Rootkit Hunter

Total=45 Periods

Lab Requirements:

S.No	Description of Equipment	Required Numbers(for batch of 30 students)
1	INTEL based desktop PC with min. 8GB RAM and 500 GB HDD, 17" or higher TFT Monitor, Keyboard and mouse	30
2	Windows 10 or higher operating system / Linux Ubuntu 20 or higher	30
3	C / C++ / Java or equivalent compiler GnuPG, Snort, N-Stalker or Equivalent.	30

Total = 45 Periods

Total Periods: (L=45; T=0; P=0) 45 Periods

Course Outcomes:**Students will be able:**

- C01 To develop code for classical Encryption Techniques to solve the problems
- C02 To build cryptosystems by applying symmetric and public key encryption algorithms
- C03 To construct code for authentication algorithms.
- C04 To develop a signature scheme using Digital signature standard
- C05 To demonstrate the network security system using open source tools

COs vs POs Mapping:

CO's	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02
C01	3	2	3	1	-	-	-	-	1	3	1	2	3	1
C02	1	3	3	3	2	-	-	-	1	2	2	2	1	2
C03	2	3	3	3	1	-	-	-	1	3	1	2	1	2
C04	3	3	1	1	1	-	-	-	3	1	1	3	2	3
C05	3	2	2	3	2	-	-	-	1	2	1	2	2	2
Average	2.4	2.6	2.4	2.2	1.5	-	-	-	1.4	2.2	1.2	2.2	1.8	2

References:

1. Build Your Own Security Lab, Michael Gregg, Wiley India